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Mechanical assessment of interfacial stability of LCP/MWCNT nanocomposites during phase transition

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ABSTRACT

Multi-walled carbon nanotubes (MWCNTs) have received considerable attention as functional additives for liquid crystalline polymers (LCPs) due to their excellent mechanical properties, light-to-heat conversion, and high compatibility with mesogenic cores. This study focuses on the effect of order-to-disorder phase transition on the mechanical properties of nanocomposites using classical molecular dynamics (MD) simulations. The results suggest that long-range π-π stacking of the aromatic rings on the nanotube surface is the key to maintaining mechanical stability under thermal loads. Thus, the MWCNT periphery effectively retards the progress of orientation field collapse with increasing temperature. The morphological consistency of the interfacial LCP allows the microstructure to preserve high load transfer efficiency even after experiencing high temperatures, improving the mechanical performance of nanocomposites.

1. Introduction

Liquid crystalline polymers (LCPs) are typically defined as polymeric materials with a high degree of order by self-organization of monomer units [1–3]. Due to the rigid rod-like structure of the mesogenic moieties, the molecules constituting the LCP spontaneously align in one-dimensional directions, resulting in anisotropy in various physical properties. One of the most interesting aspects of LCPs is the reversible phase transition from an ordered to a disordered state under a high thermal load. In the stimulated region, severe changes in microstructural anisotropy occur in conjunction with local strain. As a results, phase transitions at the molecular level lead to the contractile mechanical deformation, which can be observed even with the naked eye [4–6]. Such unique characteristics of LCPs have the potential to be applied in soft robots, artificial muscles, mechanical actuators, and smart sensors [7–15].

Recent research has been directed toward the design of LCP materials functionalized with nanofillers. In particular, multi-walled carbon nanotubes (MWCNTs) are very promising because of their excellent mechanical stability and high compatibility with the mesogenic moieties of LCP. The hexagonal lattice with sp^2 bonded carbon exhibits a high load-bearing capability against mechanical deformation, improving the strength of the overall composite system [16]. The high optical absorption of MWCNT functions as a nanoscale heat source under visible and near-infrared radiation, further increasing the applicability of LCP-based devices [17–19]. Most importantly, the slender cylindrical shape of the MWCNT aligns the adjacent liquid crystal molecules in parallel, thereby preventing the degradation of the orientational field [20–22]. Therefore, it is possible to design high-strength LCP/MWCNT nanocomposites that maintain nematic phase alignment even under extremely high nanofiller content [23].

In order to systematically design an LCP/MWCNT-based system with mechanical rigor, an analysis of the thermomechanical response, across the phase transition behavior, should be preceded. Since the origin of the thermodynamics properties is closely related to the microstructural change at the interface and the corresponding load transfer efficiency, the approaches at the sub-continuum level are required. Furthermore, interfacial binding is a crucial aspect for ensuring the mechanical reliability of nanocomposites. The microscopic distortion and rearrangement of liquid crystal molecules in contact with MWCNTs seriously compromises the sustainability of nanocomposite-based devices [24]; therefore, the microstructural effect of excessive thermal energy should be discussed in detail. However, numerous studies conducted thus far have been devote only to demonstrating the effect of the apparent deformation of nanocomposites [25–27]. Theoretical efforts to elucidate the mechanism of load transfer and the interfacial structure during the

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phase transition process have been limited to date. To design a high-strength interface, the chemical factors governing the structural stability of the system at high temperatures should also be identified.

Motivated by these considerations, the interfacial properties of the LCP/MWCNT composites are evaluated from a nanomechanical perspective. The influence of the phase change on the effective mechanical properties in the vicinity of the MWCNT surface, which has not been reported thus far, is investigated using all-atom molecular dynamics (MD) simulations. The change in the microstructure orientation under a wide range of thermal loads is quantitatively calculated. The changes in the structural properties of the interface according to the phase transition are intensively evaluated using pull-out and uniaxial tensile simulations. In particular, the stress distribution and effective deformation energy density are analyzed for each group of atoms constituting the matrix molecules. This study comprehensively provides the physicochemical principles contributing to anisotropic alignment, load transfer capability, and structural stability during the phase transition. Thus, this study is the first step in an effort to facilitate the design of phase-transition behavior in LCP-based nanocomposite structures with mechanical robustness.

2. All-atom modeling of neat LCP and LCP/MWCNT nanocomposites

Molecular modeling of the all-atom LCP/MWCNT nanocomposites and crosslinking simulations in the LCP matrix domain were performed using the commercial MD software, Materials Studio 2020 (Dassault Systèmes, France). Subsequent heating and mechanical tests were performed using the open-source code, LAMMPS (Sandia Lab., USA). The cutoff distance for noncovalent interactions was set to 12 Å and the summation of Coulomb forces was handled using the standard Ewald method with an accuracy of 10–5 kcal/mol. Thermodynamic equilibrium of the modeled system was achieved with a Nosé-Hoover thermostat and barostat. The interaction forces between the atoms constituting LCP and MWCNT and at the interface between the two materials were all calculated through the polymer consistent force field (PCFF), developed by Sun [28]. The PCFF not only secured inter- and intramolecular parameters of polymers through experiments and ab initio calculations, but also proved its applicability in various heterogeneous systems composed of polymer-metal [29], polymer-ceramic [30], and polymer-nanocarbon [31]. Also, numerous literatures have successfully reproduced the mechanical and thermodynamic behaviors of nanocarbons (**Table S1**) and liquid crystalline systems (**Table S2**) using parameters built into PCFF.

Fig. 1**(a) and (b)** show the modeled LCP and LCP/MWCNT nanocomposite systems used in the MD simulations. The preparation procedure for the unit cells was the same as that used in a previous study conducted by one of the authors [32]. The LCP network consisted of two different types of reactive monomers, 1,4-bis-[4-(3-acryloyloxypropyloxy) benzoyloxy]-2-methylbenzene (RM257) and 1,4-bis-[4-(6-acryloyloxyhexyloxy) benzoyloxy]-2-methylbenzene (RM82), in a molar ratio of 1:1. The MWCNTs were modeled as four concentric zigzag tubes composed of chiral vectors (38,0), (47,0), (56,0), and (65,0). The inner and outer diameters of the MWCNTs were 29.8 and 50.9 Å, respectively. All nanotubes were designed to have a constant length of 67.2 Å. After insertion into the unit cell box, both ends of the nanotubes were interconnected across the cell boundary. Periodic boundary conditions were imposed in all directions of the unit cell, making the nanotubes behave like cylinders of infinite length. In order to account for the interplay of LCPs with aligned MWCNTs, the liquid crystal components were initially dispersed in the same direction as the nanotubes (that is, the z-axis). The unit cell initially had a low density (0.25 g/cc) and was progressively compressed in the in-plane direction under an NPT ensemble of 1 atm at 300 K. The equations of motion for the constituent atoms were integrated using the Verlet algorithm with a time step of 1 fs. After the density of the system converged sufficiently, a crosslinking reaction occurred between the molecules. The crosslinking reaction of RM257 and RM82 was carried out through distance-based evaluation under a thermodynamic ensemble (see **Figure S1** of Supplementary Data). Acrylate groups at both ends of the reactive monomers participated in the crosslinking reaction. Relative distances between carbon atoms defined as reaction sites were calculated, and covalent bonds between the corresponding sites were newly created whenever the measured distance was shorter than a certain cutoff distance. After forming a covalent bond at each cutoff condition, both the relevant force field parameters and the covalent bond relationship were updated. For

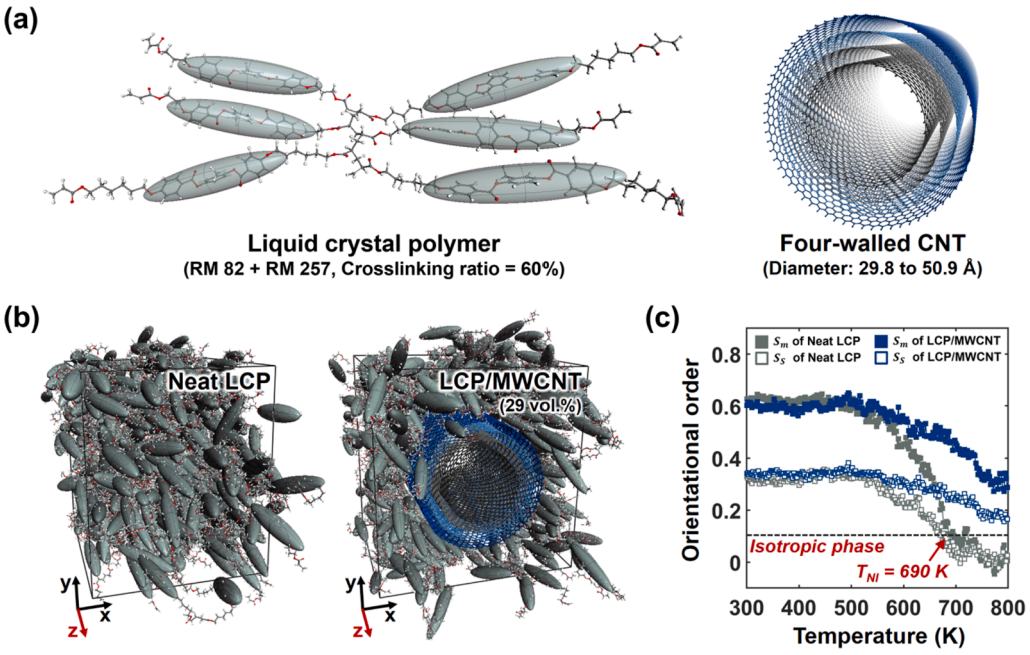


Fig. 1. (a) Molecular structures of a crosslinked liquid crystal polymer network (left) and four-walled carbon nanotubes (right). The mesogenic cores in the liquid crystal polymer network was highlighted with ellipsoidal inclusions. (b) Equilibrated microstructure of the neat LCP and LCP/MWCNT nanocomposite models with a nematic phase. (c) Thermal phase transition profiles of the models considered.

stabilization of the generated covalent bonds, a short equilibration was performed at 300 K for 50 ps under the NVT ensemble. The initially specified cutoff distance was 4.0 Å, and the value was gradually increased in 0.5 Å increments until the total crosslinking ratio of the polymer network inside the unit cell reached 60 %. Although current distance-based evaluations do not take into account the kinetic complexities of chemical reactions, it is well known that the coherent networks formed nonetheless reproduce the physical properties (e.g., morphological and structural features) of polymeric materials with high accuracy [33,34]. After the crosslinking ratio of LCP reached 60 %, the proportion of molecules participating in the reaction exceeded 80 %, which is highly consistent with the experimentally observed convergence (see **Table S3** of Supplementary Data) [35,36]. Finally, the model was further relaxed using an NVT ensemble at 300 K for 1 ns followed by an NPT ensemble at 300 K and 0.1 MPa for 1 ns. As the extremely short RMs polymerize with each other in a slightly oriented state, the final molecular structure of the LCP matrix has a three-dimensional network similar to that of thermoset polymers, while the mesogen moieties are oriented along the z-axis (see **Figure S2** for schematic of structure) [33]. Other computational modeling details can be found in a previous study [32]. After equilibration, the MWCNTs comprised 29.2 vol% of the nanocomposite system, which is consistent with the filler volume fraction of a single MWCNT sheet that can be experimentally drawn from a nanotube forest [23].

Heating-up simulations were conducted to determine the nematic-isotropic phase-transition temperature (T_{NI}) of the modeled system. From room temperature to 800 K, a constant thermal energy of 25 K/ns was applied to the neat LCP and LCP/MWCNT nanocomposite models. Simulations were performed in an NPT ensemble at 1 atm. During the heating-up period, the change in directionality of the LCP networks was evaluated using the orientational order parameter (S_r), which is defined as follows:

$$S_r = \langle \frac{3cos^2\theta_r - 1}{2} \rangle, r = mors \quad (1)$$

where θ is the angle of the liquid crystal components with the alignment direction (that is, the z-axis), and the bracket $< >$ represents the average of the derived values. The subscripts m and s are the mesogenic units and side chains that constitute the LCP, respectively. Technical details of the measurement of the angle of the liquid crystal components are shown in **Figure S3** in the **Supplementary Data**.

Fig. 1**(c)** shows the change in the orientation order parameters of the neat LCP and LCP/MWCNT nanocomposite models during the heating-up simulations. The neat LCP exhibited an orientational order parameter of 0.62 for mesogenic units at room temperature. The side chains were also affected by the alignment of the mesogenic units and showed an order parameter value of 0.31. The experimental results for the typical LCP with nematic phase ranged from 0.54 to 0.59, as determined by wide-angle X-ray scattering analysis [37,38]. Therefore, it is confirmed that the equilibrated LCP model considered in this study represents typical LCP characteristics well. Even when the temperature of the system was increased during the heating step, the molecular orientation characteristic was initially maintained. However, the order of the mesogenic units was rapidly disrupted at temperatures of 500 K or higher, indicating collapse of the initial nematic phase. This suggests that the contribution of the kinetic energy applied to the system was sufficient to break the binding force between the liquid crystal components (typically, π–π interactions of benzene rings). The profile of the order parameter continually decreased with increasing temperature and was largely scattered after complete melting. It is necessary to establish a quantitative standard to compare the thermal response properties of the materials from the simulation results. In this regard, we defined the temperature where the order parameter of mesogen unit first reaches 0.1 as the phase transition temperature. In the case of the neat LCP model, the value becomes 690 K.

The degree of alignment of the LCP/MWCNT nanocomposites at room temperature was almost identical to that of neat LCP in both the mesogenic core and side chain components. In addition, when external thermal energy was applied to the LCP/MWCNT nanocomposite system, microstructural changes started to occur at 500 K, which is consistent with the results obtained for neat LCP. However, the decay of the order parameter of the LCP/MWCNT nanocomposite was significantly retarded compared to the neat LCP. This result suggests that the thermal stability of the nematic phase of the LCPs neighboring the MWCNTs is superior to that of the neat LCP microstructures. The behavior is interpreted as the interplay between the high bending resistance of the mesogenic cores and the high aspect ratio of the nanotubes. The mesogenic cores located in the interfacial region prefer to align parallel to the nanotube axis instead of enveloping the nanotube surface to minimize the internal deformation. Consequently, well-ordered MWCNTs with a high volume fraction facilitate the alignment of LCPs and increase the resistance to distortion under thermal loading. Indeed, the high thermal stability in the interfacial region is a unique property of the LCP/MWCNT system, which has not been demonstrated in liquid crystal systems embedded with other nanofillers, such as gold nanoparticles [39].

Another interesting aspect is that the reduction in the order parameter of the LCPs in the LCP/MWCNT model was not abrupt but occurred progressively over a wide temperature range. In the neat LCP model, all polymer components were homogeneously distributed across the entire simulation box, resulting in spatially constant alignment properties. Therefore, all LCPs within the unit cell simultaneously change their alignment properties together when the temperature applied to the system reaches a certain level. In contrast, in the LCP/MWCNT nanocomposites model, the potential energy of each LCP participant was affected by the degree to which it interacts with the MWCNT. This causes the level of change in orientation to be different for each LCP molecule under thermal loading. Thus, the thermal energy distorts the alignment of liquid crystal molecules but is preferentially significant in the region far from the nanotube. Consequently, both the mesogenic units and side chains largely retained their initial orientation, even when heated above the transition temperature.

Before further consideration and analysis, it is necessary to clarify the significance and limitations of the results. The transition temperature of the LCP predicted in the current MD study (690 K) was much higher than the experimentally reported nematic-isotropic transition temperature (389–408 K) or even the melting temperature (approximately 553 K) [40–42]. The main reason for this observation is that the time scale covered by the simulation was too short to cover the actual heating and cooling rates. Nevertheless, the amount of change in the morphological properties of the microstructure before and after the phase transition follows the experimental observations very well. A comparison of the relative properties with the LCP/MWCNT nanocomposites also provided meaningful insights. Therefore, it is possible to elucidate the mechanical role of MWCNTs in the disruption process of LCP alignment and discover static aspects affecting the phase transition delay, which is the first report of its kind to the best of our knowledge. The following discussion excludes the kinetic properties of the liquid crystal field and is limited to the characterization of the interfacial structure and mechanical property changes under a given ordering state.

3. Results and discussion

3.1. Morphology of the LCP/MWCNT nanocomposites

The spatial density distribution of the LCP matrix by interaction with the MWCNT filler was evaluated. The LCP/MWCNT nanocomposites unit cell was divided into 0.1 Å thick cylindrical shells concentric with an MWCNT filler. The density of each cylindrical layer was measured individually, as shown in Fig. 2**(a)**. Note that the conditions of 0.43, 0.58, 0.72, 0.87, 1.01, and 1.16 T_{NI} indicate that the nanocomposite system was exposed to 300, 400, 500, 600, 700, and 800 K, respectively.

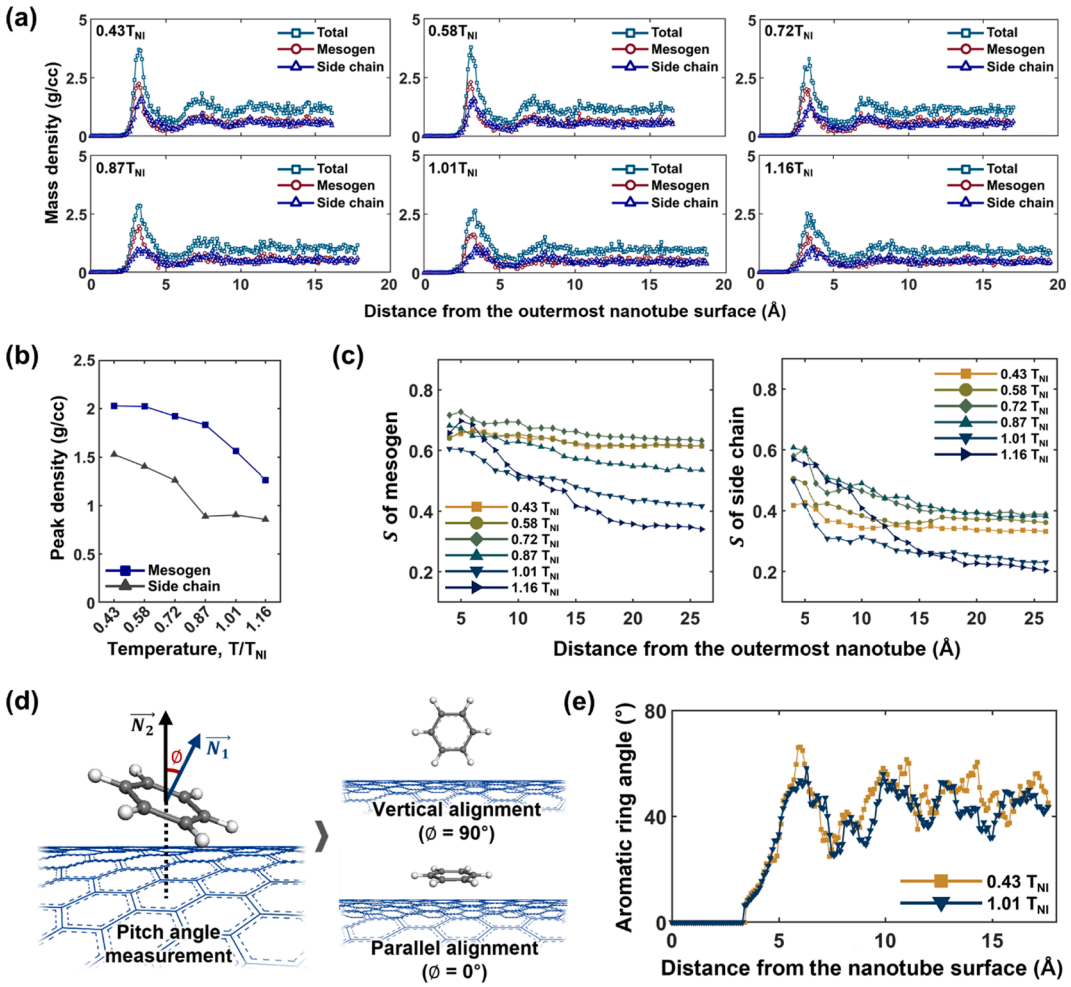


Fig. 2. Structural analysis of the LCP/MWCNT nanocomposite with increasing temperature. (a) Radial density distributions of the LCP matrix from the outermost nanotube surface. (b) Variations in the peak density of the mesogenic unit and side chains. (c) Changes in the orientational order profile (*S*) of the LCP components with temperature. (d) Conformational distribution of the aromatic ring near the MWCNT surface. The alignment property was determined from the angle formed by the normal vector of each of the aromatic rings and the nanotube surface. (e) Distributions of aromatic ring angles around the nanotube at temperatures of 0.43 and 1.01 T_{NI}.

Because the LCP network strongly interacted with the filler atoms, a dense polymer layer appeared in the immediate vicinity of the MWCNT surface. As the distance from the nanotube increased, the density profile exhibited significant oscillation and gradually converged to a bulk value (1.16 g/cc). The degree of change in the density distribution becames noticeably smaller as the temperature of the system increased. As the temperature increased from 0.43 to 1.16 T_{NI}, the peak value of the density profile decreased from 4.0 to 2.4 g/cc. Additionally, the point at which the density change relative to the distance from the nanotube disappeared shifted forward from 15 to 11 Å. Simultaneously, the converged value of the density was significantly reduced from 1.16 to 0.91 g/cc due to the distortion of the LCP alignment and the thermal expansion of the entire system.

The density profiles occupied by the mesogenic units and side chains were obtained separately to further evaluate the effect of the structural features of the matrix components on the molecular scale. The large oscillations and subsequent convergence behavior with distance from the MWCNTs were very similar for both the mesogenic units and side chains. However, the peak size of the mesogenic units was much larger than that of the side chains, and the peak position was also closer to the nanotube surface. In particular, differences between the mesogenic units and side chain relative to the peak density position were observed at all the investigated temperature conditions. The results indicated that the mesogenic units formed stronger adsorption on the MWCNT filler than the side chains, and the effect was valid within a wide temperature range.

The change in the peak density of the LCP components with increasing temperature, shown in Fig. 2**(b)**, also suggests an interesting aspect. The peak value of the mesogenic units remained large with increasing temperature but drops abruptly above the transition temperature. This behavior is distinct from the characteristics of the far field, which exhibited a linear decrease in density over the entire temperature range (see **Figure S4** in the **Supplementary Data**). Additionally, the side chains exhibited a similar trend because it is directly linked to the mesogenic units, but the magnitude of the effect was smaller than that in the mesogenic units. Particularly, the characteristics of the two LCP components were completely different when excessive heat was applied at a level that can cause phase transition. The side chains lost the noncovalent bonding force and the peak density value converged, whereas the mesogenic units had a significant residual interaction force on the nanotube surface owing to the strong π–π interactions with the MWCNTs.

Fig. 2**(c)** shows the order parameter distribution of the LCP components around the nanotube with temperature. The full data for the angle at which each molecule deviated from the nematic axis are shown in **Figure S5** of the **Supplementary Data**. At room temperature, the orientational order of both the mesogenic units and side chains increased closer to the nanotube surface. Well-aligned MWCNTs with a

high aspect ratio cause the orientation of adjacent liquid crystal molecules to be parallel to the nanotube axis, which locally enhances the orientation of the LCP network. The improvement in the directionality of these MWCNTs toward the peripheral LCPs is in good agreement with the results reported in our previous work [32].

One interesting finding is that the difference in the order parameter of the LCP along the radial direction from the nanotube surface becomes more evident as thermal energy is applied. Both the mesogenic units and side chains show a distinct change in the orientational order profile as they pass through a transition temperature, indicating a phase change in the liquid crystal field. However, in the immediate vicinity of the MWCNT surface, the alignment properties of the LCP components were rather enhanced at a high level even after the phase transition. The main cause can be found in the high interaction energy between the mesogenic units and nanotubes. The π-π binding energy formed by the nanotubes and mesogenic units was 1.5–2.8 times higher than that formed by two mesogenic units to each other [43]. Therefore, the alignment of LCPs on the MWCNT surface was maintained even when sufficient thermal energy was applied to perturb the mesogen–mesogen interaction force. Because the LCPs were homogeneously crosslinked throughout the unit cell, the anchoring effect was not localized in the interfacial region and appeared over a long range.

Additionally, the change in the order parameter with distance was more abrupt in the side chains than in the mesogenic units over the investigated temperature range. This is because the mesogenic units are not only absorbed onto the nanotube surface but also form their own binding force via intramolecular π–π interactions with their neighbors. Thus, even if the kinetic energy is increased by applying heat, a non-covalent force is present that maintain directionality. In contrast, the side chains exhibit immediate and flexible orientation degradation via torsional or flexural modes under the same environment.

The conformational distributions of the aromatic rings around the MWCNTs were further evaluated, as shown in Fig. 2**(d)**. The pitch angle is defined as the angle between the normal vector of the aromatic ring ($\overrightarrow{N_1}$) and the normal vector of the MWCNT surface ($\overrightarrow{N_2}$). Subsequently, a set of cylindrical shells of 0.1 Å thickness concentric with the nanotubes was considered, and the pitch angles associated with each layer were averaged. Fig. 2**(e)** shows the pitch angle distribution of the aromatic rings as a function of distance from the nanotube surface. The region closest to the surface of the nanotube was occupied by parallel-arranged mesogenic units, but followed by vertically arranged units. The cause that the interfacial regions are stratified with the distance from MWCNT is because the equilibrium distance of aromatic rings varies depending on the π–π stacking type [44,45]. Moreover, as the parallel aligned aromatic layer acts as an extra carbon layer on the nanotube surface, the above layered structure was repeated over a long distance. Lastly, the orientation disorder increased gradually, and the average value of the pitch angle finally converged to 45°. It is noteworthy that even above the transition temperature of 1.01 T_{NI}, the patterned structures remained almost intact. These results suggest that the effect of maintaining the initial structure even with excessive thermal energy due to the strong interaction force between the aromatic ring and nanotube surface. Specifically, the parallel-aligned mesogenic core maintained an angular distribution the same as the initial state, whereas the vertically aligned core exhibited a slight decrease in the fluctuation height of the distribution. Because aromatic rings are more strongly adsorbed to the nanotube surface when arranged in parallel [43], structural degradation caused by applied thermal energy is more severe in the case of a vertical arrangement. Nevertheless, the trend of the overall profile is preserved largely, indicating the consistency of interfacial structure by the mesogenic units.

3.2. Characterization of interfacial strength

The effect of the ordering state of the LCP matrix on interfacial strength was evaluated using the pull-out test method. Fig. 3**(a)** shows the schematic of the pull-out simulations performed in this study. The nanocomposite models equilibrated at each temperature condition were quenched to 0.1 K for 0.5 ns using the NVT ensemble. Subsequently, the boundaries of the nanocomposites were unclosed along the aligned axis. A constant velocity of 5×10^{-5} Å/fs was applied to one end of the MWCNT filler while completely restraining the surrounding LCP matrix. During separation, the variation in the interfacial energy (E_{int}) was continuously measured using the following equation:

$$E_{int} = E_{Total} - E_{MWCNT} - E_{LCP} \tag{2}$$

where E is the potential energy and the subscripts refer to each component participating in the energy calculation. Then, the pull-out force (F) was calculated as follows [46]:

$$F = \frac{\Delta E_{int}}{\Delta d} \tag{3}$$

where ΔE_{int} and Δd are the increments in the interfacial energy and nanotube displacement, respectively. To characterize the effective contribution of each molecular component constituting the polymer matrix, the entire LCP network was divided into two subgroups (that is, the mesogenic units and side chains), as shown in Fig. 3**(b)**. The grouping process was performed by linearly adding all energies allocated to the atoms of each group during the simulation. Therefore, the sum of the pull-out forces of the mesogenic units and side chains was mathematically identical to that produced by the entire system.

Herein, the pull-out test was performed under over-constrained conditions, in which the atoms constituting the LCP were completely immobilized; therefore, they could not replicate the complexity of the actual experimental environment. The primary focus of the pull-out test was to investigate the formation of non-covalent bonds between the major components constituting the LCP and the MWCNT filler when the phase transition progressed. The rearrangement or redistribution of interfacial molecules during the extraction of MWCNT was constrained under this condition; therefore, the obtained pull-out force represented only a pure morphological effect owing to the excessive thermal energy and consequent molecular disordering.

Fig. 3**(c)** shows the change in the pull-out force as a function of the displacement of the MWCNT filler (see also **Figure S6** in the **Supplementary Data** showing the change in interfacial energy). The force profile induced by the pull-out test consists of three distinct stages. Initially, a rapid increase in the pull-out force was observed because the LCP matrix prevented the displacement of the MWCNT filler (stage I). The atoms at the ends of the MWCNTs produced a cumulative force along the opposite direction of extraction in their interaction with the adjacent LCP, resulting in the resistance to separation [47,48]. Subsequently, the pull-out force profile exhibited a broad plateau at the maximum magnitude (stage II). However, the profile did not remain stable, and fluctuated significantly. This is owing to the repetitive disruption and reformation of van der Waals interactions at the nanotube surface [49]; thus, the perturbation wavelength coincides with the repeat unit length of the zigzag nanotube (see **Figure S7** in the **Supplementary Data**). Finally, as the MWCNT filler separated from the LCP matrix, the evolved pull-out force disappeared completely (stage III). Because all the LCP components are homogeneously distributed in the matrix, these three stages were also observed with the same trends in the independent profiles of the mesogen units and side chains.

Fig. 3**(d)** shows the distribution of the pull-out force applied to each nanotube shell and its contribution. Because the displacements of all shells were equal, the force distribution for each shell indicated the relative contribution of the van der Waals force of each nanotube layer to the LCP matrix. The outermost nanotube shell was found to be the source of almost all pulling forces (more than 90 %), whereas the third shell did not contribute to the interfacial bonding force. From the third shell, the interatomic distance with the LCP matrix exceeded 7 Å, where the van der Waals forces almost vanished [50]. Therefore, a force that

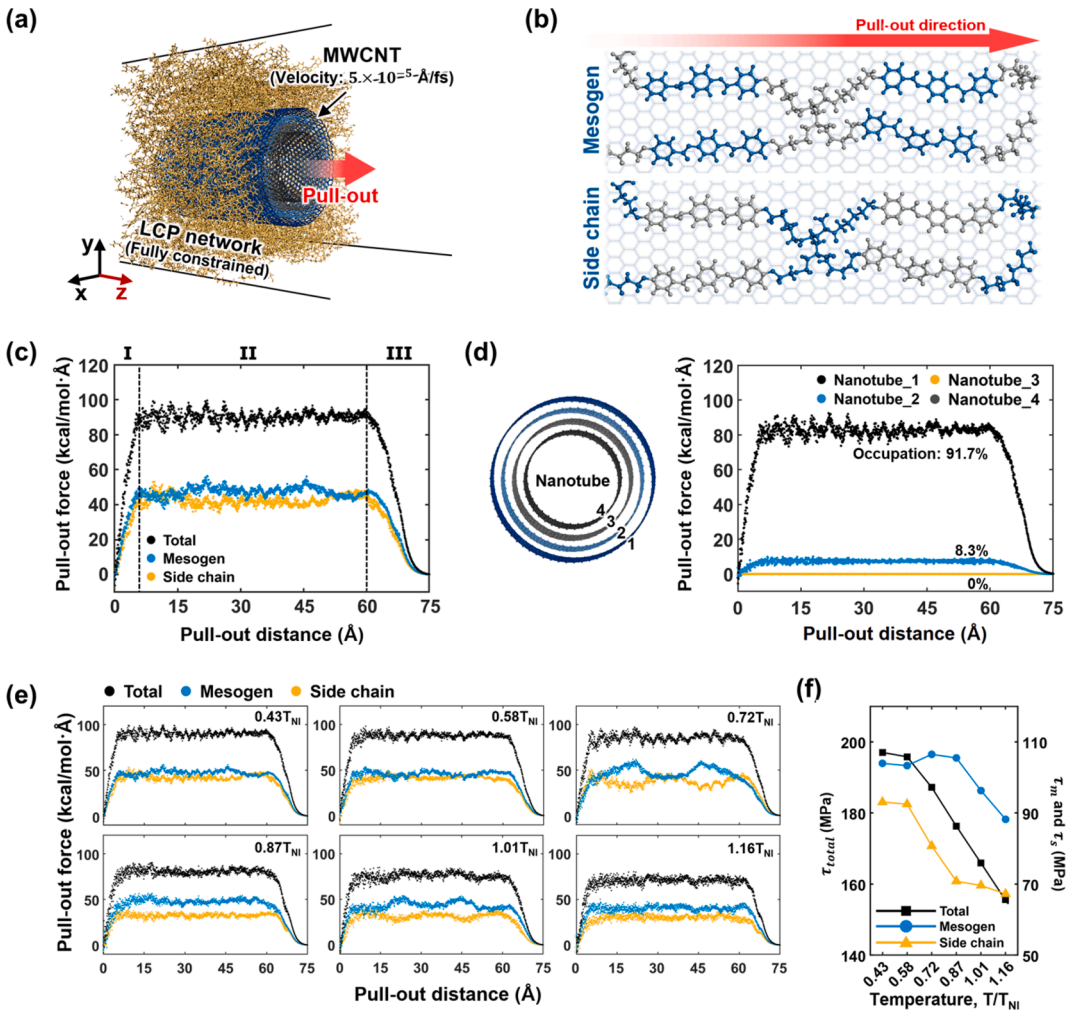


Fig. 3. Mechanical behavior at the interface of the LCP/MWCNT nanocomposite. (a) Pull-out test environment. (b) Classification of liquid crystal components for the analysis of the effective factors contributing to interfacial energy formation. (c) Profiles of the pull-out force for each liquid crystal component at a temperature of 300 K and (d) the corresponding profiles for each nanotube shell under the same conditions. (e) Changes in pull-out force with increasing temperature and (f) the corresponding interfacial shear strengths.

resists the separation of the nanotubes cannot be directly formed. However, it performs an auxiliary role in maintaining the shape of the entire MWCNT during the pullout process.

Fig. 3**(e)** shows the pull-out force distributions of the LCP components in the state in which the orientation order parameter is damaged by heat. Based on the results, two distinct trends were observed. First, the higher the temperature applied to the structure (that is, the lower the orientational order parameter of the LCP matrix), the smaller the magnitude of the pulling force. Second, the magnitude of the pulling force with MWCNTs was higher in the mesogenic units than in the side chains, and the difference was more pronounced in structures with large heat damage.

To examine the effects of temperature and LCP component type, the interfacial shear strength (τ) during stage II of the pull-out test was quantified for each component. This property was calculated using the following equation [51]:

$$\tau = \frac{F_{II}}{\pi L_{inf} D_{MWCNT}} \tag{4}$$

where F_{II} is the average pull-out force during stage II, L_{inf} is the length of the entry, and D_{MWCNT} is the outermost diameter of the MWCNT. The pull-out force was only influenced by both ends of the MWCNT filler; thus, L_{inf} was assumed to be 2 nm. As shown in Fig. 3**(f)**, the interfacial shear strength of the entire LCP decreased linearly with increasing temperature. The linearity was maintained even after the applied temperature exceeded 1.01 T_{NI} which significantly damaged the orientational order of the matrix. This finding is consistent with the results shown in Fig. 2, in which the morphological change caused by temperature occurred predominantly in the far field instead of at the MWCNT interface. However, the behaviors of the mesogenic units and side chains constituting the LCP matrix were significantly different from each other. While the mesogenic units retained their initial strength below the transition temperature, the strength of the side chain quickly collapses from 0.58 T_{NI} and converges to a specific value.

Again, the outstanding structural stability of the mesogenic group can be explained by forcibly improved crystallinity in the interfacial region with the MWCNTs. The liquid crystal molecules oriented along the nanotube axis largely retained their alignment field despite the disturbance of the regions outside the interface. Therefore, the interfacial shear strength of the mesogenic units was barely affected by the temperature change until the system underwent a phase transition. Even after the phase transition occurred, the reduction in strength exhibited by the mesogenic units was minimal compared with that of the side chains. In contrast, an instantaneous reduction in strength was observed with temperature for the side chains, which is typical for amorphous polymer-CNT systems [52,53]. The bonding force between the untreated nanotube surface and hydrocarbon chains was very weak and exhibited low mechanical resistance to shear forces. The weak bonding force to the

nanotube surface causes easy displacement of the neighboring side chains, even with a slight increase in temperature, indicating a low effective strength. Therefore, as the system approaches the transition temperature, the side chains in the interfacial region completely lose the bonding force induced by the co-alignment with the nanotubes.

3.3. Evaluation of mechanical properties

The effect of molecular-level disorder formed by the application of heat on the mechanical properties was evaluated using the tensile test method. In the process of heating the LCP and LCP/MWCNT nanocomposite models from 0.43 to 1.16 T_{NI}, each equilibrated morphology was obtained at a temperature interval of 100 K. Subsequently, a temperature of 0.1 K was applied for 500 ps using the NVT ensemble, thereby eliminating all interference caused by the thermal effect possessed by atoms. Therefore, the mechanical properties discussed in this section are not intended to describe the behavior of the material under the corresponding temperature conditions. Rather, it is to extract the pure structural effect of the orientational order distribution reformed by the application of heat.

Uniaxial tensile strain of up to 0.1 % was applied to each prepared model at a strain rate of 10^6/s. A cryogenic temperature (0.1 K) was maintained using the NVT ensemble during the tensile simulation. As the positions and velocities of the atoms are updated by the elongation of the simulation box, the virial stress of the atoms constituting the system were calculated as follows [54]:

$$\boldsymbol{\sigma} = \frac{1}{V}\left(-\sum_i^N m_i(\boldsymbol{v}_i\boldsymbol{v}_i^T) + \frac{1}{2}\sum_i^N\sum_{j\neq i}^N \boldsymbol{r}_{ij}\boldsymbol{F}_{ij}\right) \tag{5}$$

where V is the volume, N is the number of constituent atoms, m_i and $\boldsymbol{v}_i$ are the mass and velocity of atom i, respectively. Additionally, $\boldsymbol{r}_{ij}$ and $\boldsymbol{F}_{ij}$ are the relative distance and interaction force between atoms i and j, respectively. Because a very low temperature is applied, the first (kinetic) term is negligible; therefore, only the change in the second (potential) energy formed by the atoms dominates the stress in the system. Each stiffness component of the system is written in Voigt notation as follows:

$$\sigma_i = C_{ij}\varepsilon_j (i,j = 1,\cdots,6) \tag{6}$$

where ε_j is the unit cell strain. Both the LCP and LCP/MWCNT nanocomposites were transversely isotropic, and the axial stiffness components were obtained from the following relationship.

$C_{11} = \frac{(C_{xx}+C_{yy})}{2}$ and $C_{33} = C_{zz}$ (7).

The contribution of each molecule to the mechanical stress formation of the entire system was quantitatively analyzed according to our previously reported methodology [55,56]. Each effective stress tensor generated in the MWCNTs, mesogens, and side chains was obtained individually by confining the atoms considered in Equation (5) to only the relevant components. The volume occupied by each component was calculated as the spatial area of the atom defined by Voronoi tessellation, as listed in **Table S4** in the **Supplementary Data**. Subsequently, the substantial work done by the nanocomposite components during mechanical tension was derived using the effective deformation energy density as follows:

$$u_k = \int_0^{\varepsilon_0} \sigma_{xx}^k d\varepsilon \tag{8}$$

where σ_{xx}^k and ε_0 are the atomic stresses of component k and the total strain, respectively. Thus, the deformation energy density of the entire system is equal to the linear sum of the energies distributed to the nanocomposite component as follows:

$$u = \sum_k u_k \varphi_k \tag{9}$$

where φ_k is the volume fraction of component k.

Note that the present tensile simulation is not to reproduce the actual experimental test, but rather as a conceptual framework to numerically reveal the structural anisotropy represented by the interfacial region of LCP/MWCNT. Because MWCNTs are interconnected across periodic boundaries and are infinite in length, all layers of the nanotube stretch simultaneously when evaluating their longitudinal properties. Therefore, differences in the strain field due to asymmetric load transfer between layers of nanotubes are not considered in this simulation. Instead, we can analyze the change in the deformation energy density distribution constituting the interface region for each tensile direction in a static environment. In other words, the morphological changes and structural anisotropy of the interfacial region during the phase transition can be explicitly characterized from a mechanical point of view.

Fig. 4**(a)** shows the change in mechanical stiffness of the neat LCP as the arrangement was disrupted by heat. Before the phase transition, the stiffness in the axial direction was more than twice that in the transverse direction, which represents the anisotropy of typical liquid crystal materials. As higher temperatures are experienced, the magnitude of the stiffness component of the microstructure decreases significantly in both directions. The anisotropy (C_{33}/C_{11}) rather increases until the disorder occurs, because the transverse stiffness component is more sensitive to changes in non-covalent force due to temperature (see **Figure S8** in the **Supplementary Data**). As it undergoes a phase disorder, the transverse stiffness is largely maintained due to the reorientation of the alignment, whereas the axial stiffness continued to decrease. Therefore, the stiffness ratio of the LCP approached one, indicating the isotropic phase of the microstructure.

Fig. 4**(b)** shows the elastic properties of the LCP/MWCNT nanocomposite and the deformation energy density assigned to each component. Not only the mechanical stiffness of the nanocomposites was higher compared to the neat LCP material in all directions, but the strengthening effect was preserved in all temperature conditions. Because the MWCNTs considered in the model have an infinite length, almost all energy increases that occur during the application of an external load in the longitudinal direction are achieved by the direct stretching of the MWCNTs (the elastic stiffness is 25.8 times that of the neat LCP model). However, the collapse of the alignment of the matrix due to heat has a significant effect to reduce the axial stiffness linearly. The decrease in the filler volume fraction of the nanocomposite due to the thermal expansion of the LCP matrix is the main cause of the reduction in the mechanical properties (see **Table S4**). Because MWCNTs have no substantial structural collapse or decomposition, even after the supply of thermal energy, the independent deformation energy under external loads is almost constant, regardless of the temperature applied.

The transverse stiffness of the LCP/MWCNT decreased exponentially with temperature, similar to that of the neat LCP, and subsequently converged after the phase transition. The magnitude of the stiffness component at room temperature was 56.6 % higher than that of the neat LCP, indicating that MWCNTs functioned as reinforcing materials even in the transverse direction. The abundant π–π interactions around the MWCNTs lead to the high level of structural stability, and thus, a high load-transfer efficiency was achieved. Notably, these reinforcing effects were still observed even in the structures that had undergone a phase transition. The deformation energy density analysis revealed that the MWCNTs made a highest contribution even after a nematic to isotropic phase change occurred in the system subjected to lateral loading.

The origin of the effective transversal load transfer to the structure after high thermal loading was clearly identified based on the deformation energy distributed in the matrix region, as shown in Fig. 4**(c)**. Regardless of the thermal energy applied, all deformation energy profiles exhibited distinctly high values near the MWCNT filler, which indicates the high deformability of the interfacial LCP by an external load [57,58]. As previously discussed, the crystallinity of the LCPs in the contact region with the MWCNTs was forcibly elevated, significantly enhancing the noncovalent interaction force between the mesogenic

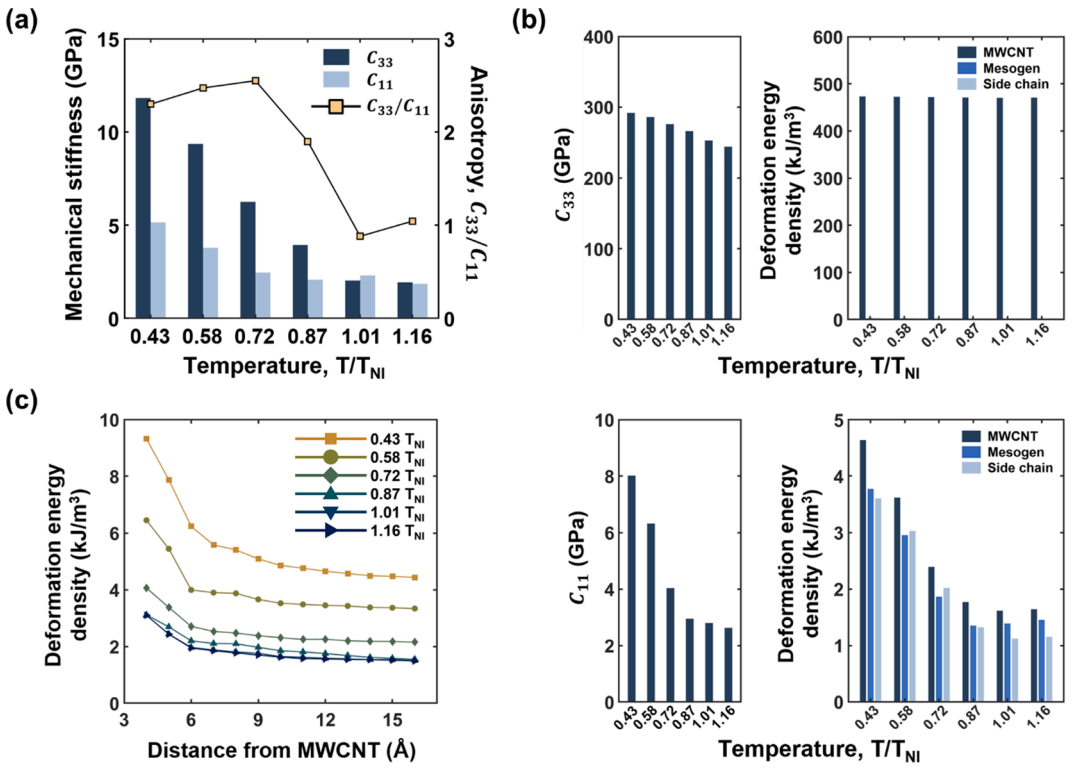


Fig. 4. Mechanical properties based on temperature exposure. (a) Stiffness components of the neat LCP model and their anisotropy. (b) Elastic stiffness of the LCP/MWCNT nanocomposite and deformation energy density allocated to each nanocomposite component. The results were presented for the applied mechanical load in the axial direction (top) and transverse direction (bottom). (c) Distributions of the deformation energy density in the LCP matrix during transverse loading of the nanocomposite.

units. Thus, the LCPs in the region adjacent to the MWCNTs behave more collectively under external loads. When transverse strain is applied, the LCPs stacked on the MWCNT surface are deformed together, which causes both high deformability of the interface and storage of a large amount of elastic energy.

As the nanocomposite models were exposed to high temperatures, the deformation energy profile progressively decreased due to the mechanical weakening of the LCP matrix. The profiles almost converged upon phase transition. Still, the LCPs adjacent to the MWCNT surface exhibited distinct differences from those in the far-field region. Regardless of the temperature experienced, the maximum value of the deformation energy density always occurred in the region closest to the MWCNTs, which was twice as large as the far field. This result suggests that the interaction force produced by the MWCNTs is effective, even under sufficiently high thermal energy to perturb the alignment of the LCP matrix. This finding is also in line with the morphological consistency of the mesogenic groups in the interfacial region over a wide temperature range, which substantially enhances the mechanical properties of materials, even after the phase transition.

4. Conclusions

In this study, the interfacial properties of LCP/MWCNT nanocomposites were characterized from a nanomechanical perspective. The correlation between thermal loading, microstructural disorder, and mechanical properties were described in terms of the progression of the phase transition.

The results suggest that π-π interactions between the series-connected aromatic rings and carbon nanotubes are a key in maintaining high anisotropic stability under thermal loads. The anchoring effect induced by MWCNTs not only aligned the adjacent liquid crystal molecules along the nanotube axis, but also significantly retarded the collapse of the orientation field of the nanocomposite. Because the enhancement of thermal stability is inversely proportional to the distance from the MWCNTs, the alignment disorder was sensitive only in the far field. For the LCP molecules in the interfacial region, each component of the mesogen units and side chains exhibited different behaviors. The mesogens forcibly immobilized on the surface of the MWCNTs retained their initial morphological features, even when excessive thermal energy was applied at a level sufficient to disrupt the intermolecular interactions. Whereas, the side chains connected to the mesogenic cores exhibits immediate and severe structural deterioration with a slight temperature rise.

The structural change at the molecular level due to the phase transition was also confirmed as the same tendency in the analysis of mechanical properties. The mesogenic core dominated the interfacial shear strength over all temperature ranges, and thus the mechanical properties and anisotropy at the interface could be maintained at a significant level even after the LCP/MWCNT nanocomposites experienced high temperatures. In contrast, the side chains show an immediate mechanical degradation even with insignificant thermal energy, so exhibit only minor contribution to interfacial stability. Finally, the morphological consistency of mesogen was revealed to lead to the high load transfer efficiency across the nanocomposites interface under mechanical loading.

The present model can serve as a milestone in understanding the molecular-level physicochemical factors and their operating mechanisms required to design intelligent LCP nanocomposites with high strength and high anisotropy. With these results as a first step, the model needs to be refined in the future to understand the physicochemical factors experienced by LCP/MWCNTs in the synthesis stage and their behavior in an environment where they are subjected to actual mechanical loads. It is of particular urgency to elucidate the effects of crosslinking between nanotube layers [59], discontinuities in nanotubes of finite length [60] and the waviness and alignment quality of nanotubes [46] on their interaction with LCPs. In order to simulate the

realistic behavior at the LCP/MWCNT interface, it is also important to extend the model considering the dynamic properties of the material, such as the slip between nanotube walls [61,62] and the discontinuous strain field at the interface. Chemical treatment and improvement of load transfer efficiency by interfacial crosslinking will also be dealt in the near future [63–65]. The expansion of the coverage of the model ultimately enables the design of mechanical properties of LCP-based nanocomposites that are optimally suited to the user environment. The diversification and specialization of the next-generation LCP-based intelligent structure design based on the rationale suggested by the model is also underway.

CRediT authorship contribution statement

Hongdeok Kim: Methodology, Software, Visualization, Writing – original draft. **Joonmyung Choi:** Conceptualization, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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Appendix A. Supplementary data

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References

[1] Van Oosten CL, Bastiaansen CW, Broer DJ. Printed artificial cilia from liquid-crystal network actuators modularly driven by light. Nat Mater 2009;8(8):677–82.
[2] Yang Y, Zhan W, Peng R, He C, Pang X, Shi D, et al. Graphene-enabled superior and tunable photomechanical actuation in liquid crystalline elastomer nanocomposites. Adv Mater 2015;27(41):6376–81.
[3] Shahsavan H, Yu L, Jákli A, Zhao B. Smart biomimetic micro/nanostructures based on liquid crystal elastomers and networks. Soft Matter 2017;13(44):8006–22.
[4] de Haan LT, Sánchez-Somolinos C, Bastiaansen CM, Schenning AP, Broer DJ. Engineering of complex order and the macroscopic deformation of liquid crystal polymer networks. Angew Chem 2012;124(50):12637–40.
[5] Ge F, Zhao Y. Microstructured actuation of liquid crystal polymer networks. Adv Funct Mater 2020;30(2):1901890.
[6] Lv J-a, Liu Y, Wei J, Chen E, Qin L, Yu Y. Photocontrol of fluid slugs in liquid crystal polymer microactuators. Nature 2016;537(7619):179–84.
[7] Xiao YY, Jiang ZC, Tong X, Zhao Y. Biomimetic locomotion of electrically powered "Janus" soft robots using a liquid crystal polymer. Adv Mater 2019;31(36):1903452.
[8] Liu J, Gao Y, Wang H, Poling-Skutvik R, Osuji CO, Yang S. Shaping and locomotion of soft robots using filament actuators made from liquid crystal elastomer–carbon nanotube composites. Adv Intell Syst 2020;2(6):1900163.
[9] Li S, Bai H, Liu Z, Zhang X, Huang C, Wiesner LW, et al. Digital light processing of liquid crystal elastomers for self-sensing artificial muscles. Sci Adv 2021;7(30).
[10] Kim H, Ahn S-k, Mackie DM, Kwon J, Kim SH, Choi C, et al. Shape morphing smart 3D actuator materials for micro soft robot. Mater Today 2020;41:243–69.
[11] Kotikian A, Morales JM, Lu A, Mueller J, Davidson ZS, Boley JW, et al. Innervated, Self-Sensing Liquid Crystal Elastomer Actuators with Closed Loop Control. Adv Mater 2021;33(27):2101814.
[12] Kim H, Gibson J, Maeng J, Saed MO, Pimentel K, Rashed P, et al. Responsive, 3D electronics enabled by liquid crystal elastomer substrates. ACS Appl Mater Interfaces 2019;11(21):19506–13.
[13] de Haan LT, Gimenez-Pinto V, Konya A, Nguyen TS, Verjans JM, Sánchez-Somolinos C, et al. Accordion-like actuators of multiple 3D patterned liquid crystal polymer films. Adv Funct Mater 2014;24(9):1251–8.
[14] Ge F, Yang R, Tong X, Camerel F, Zhao Y. A multifunctional dye-doped liquid crystal polymer actuator: light-guided transportation, turning in locomotion, and autonomous motion. Angew Chem 2018;130(36):11932–7.
[15] Li C, Liu Y, Huang X, Jiang H. Direct sun-driven artificial heliotropism for solar energy harvesting based on a photo-thermomechanical liquid-crystal elastomer nanocomposite. Adv Funct Mater 2012;22(24):5166–74.
[16] Jang HG, Yang B, Khil M-S, Kim SY, Kim J. Comprehensive study of effects of filler length on mechanical, electrical, and thermal properties of multi-walled carbon nanotube/polyamide 6 composites. Compos A Appl Sci Manuf 2019;125:105542.
[17] Han NR, Cho JW. Effect of click coupled hybrids of graphene oxide and thin-walled carbon nanotubes on the mechanical properties of polyurethane nanocomposites. Compos A Appl Sci Manuf 2018;109:376–81.
[18] Wang F, Wang W, Zhang C, Tang J, Zeng X, Wan X. Scalable manufactured bio-based polymer nanocomposite with instantaneous near-infrared light-actuated targeted shape memory and remote-controlled accurate self-healing. Compos B Eng 2021;219:108927.
[19] Yu Y, Li L, Liu E, Han X, Wang J, Xie Y, et al. Light-driven core-shell fiber actuator based on carbon nanotubes/liquid crystal elastomer for artificial muscle and phototropic locomotion. Carbon 2022;187:97–107.
[20] Ji Y, Fan F, Zhang Z, Cheng J, Chang S. Active terahertz liquid crystal device with carbon nanotube film as both alignment layer and transparent electrodes. Carbon 2022;190:376–83.
[21] Jiang K, Wang J, Li Q, Liu L, Liu C, Fan S. Superaligned carbon nanotube arrays, films, and yarns: a road to applications. Adv Mater 2011;23(9):1154–61.
[22] Moon J, Choi J, Cho M. Opto-mechanical behavior and interfacial characteristics of crosslinked liquid crystalline polymer composites with carbon nanotube fillers. Carbon 2017;121:181–92.
[23] Kim H, Lee JA, Ambulo CP, Lee HB, Kim SH, Naik VV, et al. Intelligently actuating liquid crystal elastomer-carbon nanotube composites. Adv Funct Mater 2019;29(48):1905063.
[24] Chen L, Dong Y, Tang C-Y, Zhong L, Law W-C, Tsui GCP, et al. Xie X. Development of direct-laser-printable light-powered nanocomposites. ACS Appl Mater Interfaces 2019;11(21):19541–53.
[25] Yang L, Setyowati K, Li A, Gong S, Chen J. Reversible infrared actuation of carbon nanotube–liquid crystalline elastomer nanocomposites. Adv Mater 2008;20(12):2271–5.
[26] Wang M, Sayed SM, Guo L-X, Lin B-P, Zhang X-Q, Sun Y, et al. Multi-stimuli responsive carbon nanotube incorporated polysiloxane azobenzene liquid crystalline elastomer composites. Macromolecules 2016;49(2):663–71.
[27] Ahn C, Li K, Cai S. Light or thermally powered autonomous rolling of an elastomer rod. ACS Appl Mater Interfaces 2018;10(30):25689–96.
[28] Sun H. Force field for computation of conformational energies, structures, and vibrational frequencies of aromatic polyesters. J Comput Chem 1994;15(7):752–68.
[29] Choi J, Shin H, Cho M. Multiscale multiphysical analysis of photo-mechanical properties of interphase in light-responsive polymer nanocomposites. Compos Sci Technol 2018;160:32–41.
[30] Kim B, Choi J, Yang S, Yu S, Cho M. Influence of crosslink density on the interfacial characteristics of epoxy nanocomposites. Polymer 2015;60:186–97.
[31] Yang S, Shin H, Cho M. Contribution of oxygen functional groups in graphene to the mechanical and interfacial behaviour of nanocomposites: Molecular dynamics and micromechanics study. Int J Mech Sci 2021;189:105972.
[32] Kim H, Kim H, Choi J. Interface mechanics of liquid crystal polymer nanocomposites with high concentrations of MWCNTs. Compos Sci Technol 2022;222:109376.
[33] Chowdhury SC, Elder RM, Sirk TW, Gillespie JW. Epoxy resin thermo-mechanics and failure modes: Effects of cure and cross-linker length. Compos B Eng 2020;186:107814.
[34] Unger R, Braun U, Fankhanel J, Daum B, Azash B, Rolfes R. Molecular modelling of epoxy resin crosslinking experimentally validated by near-infrared spectroscopy. Comput Mater Sci 2019;161:223–35.
[35] Broer DJ, Boven J, Mol GN, Challa G. In-situ photopolymerization of oriented liquid-crystalline acrylates, 3. Oriented polymer networks from a mesogenic diacrylate. Makromol Chem 1989;190:2255–68.
[36] Kamal T, Park S. Shape-responsive actuator from a single layer of a liquid-crystal polymer. ACS Appl Mater Interfaces 2014;6(20):18048–54.
[37] Traugutt N, Volpe R, Bollinger M, Saed MO, Torbati AH, Yu K, et al. Liquid-crystal order during synthesis affects main-chain liquid-crystal elastomer behavior. Soft Matter 2017;13(39):7013–25.
[38] Luo F, Yang S, Yan P, Li H, Huang B, Qian Q, et al. Orientation behavior and thermal conductivity of liquid crystal polymer composites based on Three-Dimensional printing. Compos A Appl Sci Manuf 2022;160:107059.
[39] De Jesús-Téllez MA, Felix-Serrano I, Rodríguez-González RJ, Navarro-Rodríguez D, Larios-López L. Effect of gold and graphene oxide nanoparticles on the thermo-and photo-actuation of monodomain liquid crystal elastomers. Polymer 2020;205:122837.
[40] Liu D, Broer DJ. Liquid crystal polymer networks: preparation, properties, and applications of films with patterned molecular alignment. Langmuir 2014;30(45):13499–509.
[41] Chen T, Mansfield CD, Ju L, Baird DG. The influence of mechanical recycling on the properties of thermotropic liquid crystalline polymer and long glass fiber reinforced polypropylene. Compos B Eng 2020;200:108316.

[42] Xu F, Xin Y, Li T. Tribological enhancement effect of main-chain thermotropic liquid crystalline polymer. Compos A Appl Sci Manuf 2018;108:69–78.
[43] Yang Z, Wang Z, Tian X, Xiu P, Zhou R. Amino acid analogues bind to carbon nanotube via π-π interactions: Comparison of molecular mechanical and quantum mechanical calculations. J Chem Phys 2012;136(2):01B607.
[44] Hohenstein EG, Sherrill CD. Effects of heteroatoms on aromatic π– π interactions: benzene– pyridine and pyridine dimer. J Phys Chem A 2009;113(5):878–86.
[45] Sinnokrot MO, Sherrill CD. Highly accurate coupled cluster potential energy curves for the benzene dimer: sandwich, T-shaped, and parallel-displaced configurations. J Phys Chem A 2004;108(46):10200–7.
[46] Alian A, Meguid S. Molecular dynamics simulations of the effect of waviness and agglomeration of CNTs on interface strength of thermoset nanocomposites. Phys Chem Chem Phys 2017;19(6):4426–34.
[47] Duan K, Li L, Hu Y, Wang X. Interface mechanical properties of graphene reinforced copper nanocomposites. Mater Res Express 2017;4(11):115020.
[48] Wernik J, Cornwell-Mott B, Meguid S. Determination of the interfacial properties of carbon nanotube reinforced polymer composites using atomistic-based continuum model. Int J Solids Struct 2012;49(13):1852–63.
[49] Duan Ke, Li Li, Wang F, Liu S, Hu Y, Wang X. New insights into interface interactions of CNT-reinforced epoxy nanocomposites. Compos Sci Technol 2021;204:108638.
[50] Duan K, Li L, Wang F, Meng W, Hu Y, Wang X. Importance of interface in the coarse-grained model of CNT/epoxy nanocomposites. Nanomaterials 2019;9(10):1479.
[51] Chawla R, Sharma S. Molecular dynamics simulation of carbon nanotube pull-out from polyethylene matrix. Compos Sci Technol 2017;144:169–77.
[52] Guru K, Mishra S, Shukla K. Effect of temperature and functionalization on the interfacial properties of CNT reinforced nanocomposites. Appl Surf Sci 2015;349:59–65.
[53] Singh A, Kumar D. Temperature effects on the interfacial behavior of functionalized carbon nanotube–polyethylene nanocomposite using molecular dynamics simulation. Proc Inst Mech Eng N: J Nanomater Nanoeng Nanosyst 2019;233(1):3–15.
[54] Heyes DM. Pressure tensor of partial-charge and point-dipole lattices with bulk and surface geometries. Phys Rev B 1994;49(2):755–64.
[55] Kim H, Choi J. Subcontinuum Interpretation of Mechanical Behavior for Cross-Linked Epoxy Networks. Macromolecules 2022;55(14):5916–25.
[56] Kim H, Choi J. Interfacial and mechanical properties of liquid crystalline elastomer nanocomposites with grafted Au nanoparticles: A molecular dynamics study. Polymer 2021;218:123525.
[57] Choi J, Shin H, Cho M. A multiscale mechanical model for the effective interphase of SWNT/epoxy nanocomposite. Polymer 2016;89:159–71.
[58] Choi J, Shin H, Yang S, Cho M. The influence of nanoparticle size on the mechanical properties of polymer nanocomposites and the associated interphase region: A multiscale approach. Compos Struct 2015;119:365–76.
[59] Xiang Y, Shimoyama K, Shirasu K, Yamamoto G. Machine Learning-Assisted High-Throughput Molecular Dynamics Simulation of High-Mechanical Performance Carbon Nanotube Structure. Nanomaterials 2020;10(12):2459.
[60] Jensen BD, Odegard GM, Kim JW, Sauti G, Siochi EJ, Wise KE. Simulating the effects of carbon nanotube continuity and interfacial bonding on composite strength and stiffness. Compos Sci Technol 2018;166(29):10–9.
[61] Li Y, Hu N, Yamamoto G, Wang Z, Hashida T, Asanuma H, et al. Molecular mechanics simulation of the sliding behavior between nested walls in a multi-walled carbon nanotube. Carbon 2010;48(10):2934–40.
[62] Yamamoto G, Liu S, Hu N, Hashida T, Liu Y, Yan C, et al. Prediction of pull-out force of multi-walled carbon nanotube (MWCNT) in sword-in-sheath mode. Comput Mater Sci 2012;60:7–12.
[63] Zhang J, Sun D, Zhang B, Sun Q, Zhang Y, Liu S, et al. Intrinsic carbon nanotube liquid crystalline elastomer photoactuators for high-definition biomechanics. Mater Horiz 2022;9(3):1045–56.
[64] Chowdhury SC, Okabe T. Computer simulation of carbon nanotube pull-out from polymer by the molecular dynamics method. Compos A Appl Sci Manuf 2007;38(3):747–54.
[65] Chowdhury SC, Okabe T, Nishikawa M. Effects of Vacancy Defects on the Interfacial Shear Strength of Carbon Nanotube Reinforced Polymer Composite. J Nanosci Nanotechnol 2010;10(2):739–45.